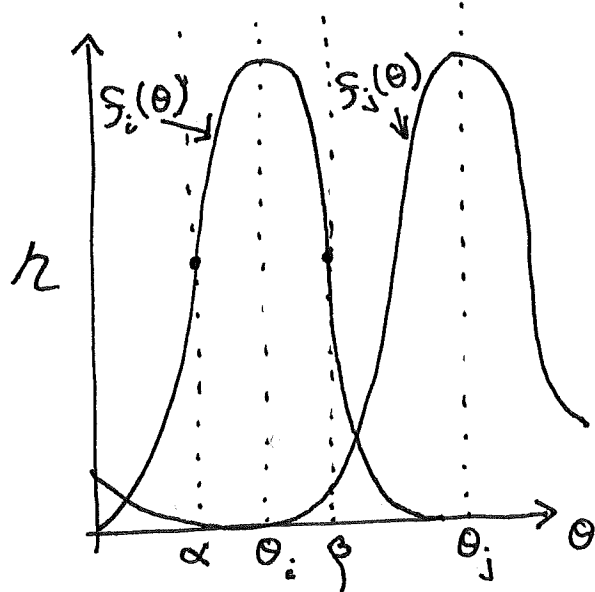


Neural population decoding

Back to MT: single neurons are ambiguous.



$$P_i(k|\theta) = \frac{(f_i(\theta)\Delta t)^k e^{-f_i(\theta)\Delta t}}{k!}$$

$$P_i(k|\alpha) = P_i(k|\beta)$$

Population breaks degeneracy

$$P(k_i, k_j|\alpha) \neq P(k_i, k_j|\beta)$$

Population of $N \gg 1$ independent Poisson neurons

$$\underbrace{\theta_i = \frac{2\pi i}{N}}_{\text{Uniformly spaced preferred directions}}; \underbrace{f_i(\theta) = f(\theta - \theta_i)}_{\text{Same tuning curve shape, different } \theta_i}; \underbrace{P(k_1, \dots, k_N|\theta) = \prod_{i=1}^N P_i(k_i|\theta)}_{\text{Independent neurons}}$$

Likelihood function:

$$L(\theta) = P(k_1, \dots, k_N|\theta) = \prod_{i=1}^N \frac{(f_i(\theta)T)^{k_i} e^{-f_i(\theta)T}}{k_i!}; \quad T: \text{trial duration}$$

Log-likelihood function:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^N \left(\underbrace{k_i \log(f_i(\theta)T)}_{\text{Linear in } k_i} - \underbrace{f_i(\theta)T - \log k_i!}_{\theta\text{-independent}} \right)$$

Asymptotic decoding performance of maximum likelihood is set by Fisher information. Need to calculate second derivative.

$$\begin{aligned} \frac{d\ell(\theta)}{d\theta} &= \sum_{i=1}^N \left(\frac{k_i}{f_i(\theta)T} \frac{df_i(\theta)}{d\theta} T - \frac{df_i(\theta)}{d\theta} T \right) \\ &= \sum_{i=1}^N \frac{df_i(\theta)}{d\theta} T \left(\frac{k_i}{f_i(\theta)T} - 1 \right) \end{aligned}$$

$$\begin{aligned} \frac{d^2 \ell(\theta)}{d\theta^2} &= \sum_{i=1}^N \left(\frac{d^2 \mathcal{F}_i(\theta)}{d\theta^2} T \left(\frac{k_i}{\mathcal{F}_i(\theta)T} - 1 \right) + \frac{d\mathcal{F}_i(\theta)}{d\theta} T \left(-\frac{k_i}{(\mathcal{F}_i(\theta)T)^2} \right) \right. \\ &\quad \left. + \frac{d\mathcal{F}_i(\theta)}{d\theta} T \right) \\ &= \sum_{i=1}^N \left(\frac{d^2 \mathcal{F}_i(\theta)}{d\theta^2} T \left(\frac{k_i}{\mathcal{F}_i(\theta)T} - 1 \right) - \left(\frac{d\mathcal{F}_i(\theta)}{d\theta} \right)^2 \frac{k_i}{(\mathcal{F}_i(\theta))^2} \right) \end{aligned}$$

Recall that

$$\langle k_i \rangle_{k|\theta} = \mathcal{F}_i(\theta)T$$

Thus,

$$J(\theta) = - \left\langle \frac{d^2 \ell(\theta)}{d\theta^2} \right\rangle_{k|\theta} = \sum_{i=1}^N \frac{(d\mathcal{F}_i(\theta)/d\theta)^2}{\mathcal{F}_i(\theta)} T$$

For large N , we can approximate sum with integral.

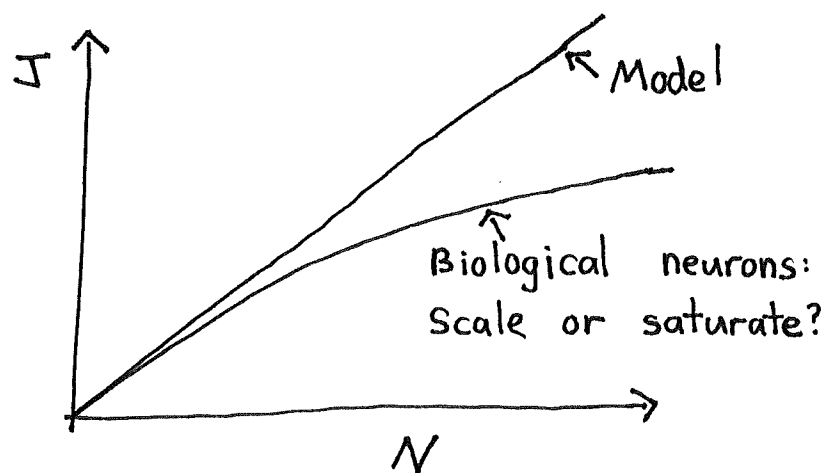
$$\sum_{i=1}^N F(\theta_i) = \sum_{i=1}^N \frac{\Delta\theta_i}{2\pi/N} F(\theta_i) \xrightarrow{N \gg 1} N \int_0^{2\pi} \frac{d\theta_i}{2\pi} F(\theta_i)$$

The Fisher information becomes

$$\begin{aligned} J(\theta) &= NT \int_0^{2\pi} \frac{d\theta_i}{2\pi} \frac{(d\mathcal{F}(\theta - \theta_i)/d\theta)^2}{\mathcal{F}(\theta - \theta_i)} \\ &= \underbrace{NT}_{\text{Scaling}} \underbrace{\int_0^{2\pi} \frac{d\theta'}{2\pi} \frac{(d\mathcal{F}(\theta')/d\theta)^2}{\mathcal{F}(\theta')}}_{\text{Tuning curve dependence}} \end{aligned}$$

The decoder works arbitrarily well for large systems.

$$\hat{\theta} \sim \mathcal{N}(\theta_0, 1/J(\theta_0))$$

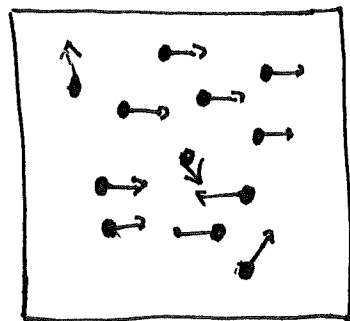


Why is this issue important?

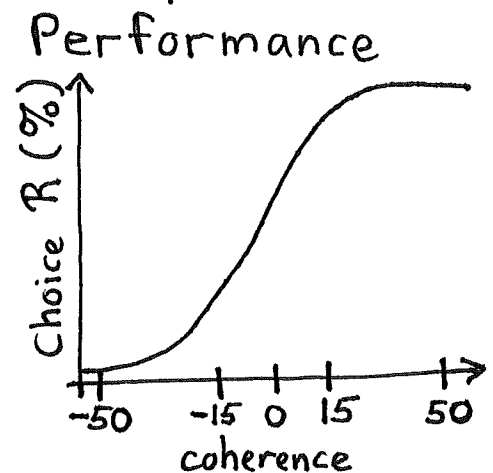
1. Fisher information limits any decoder, not just maximum likelihood. For unbiased decoders,

$$\underbrace{\langle \hat{\theta} \rangle = \theta_0}_{\text{Unbiased}}, \quad \underbrace{\text{Var}(\hat{\theta}) \geq \frac{1}{J(\theta)}}_{\text{Cramer-Rao Lower Bound}}$$

2. Perceptual errors are substantial



Behavior
L or R?

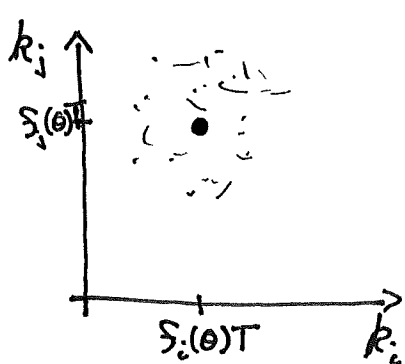


3. Neurons are not independent

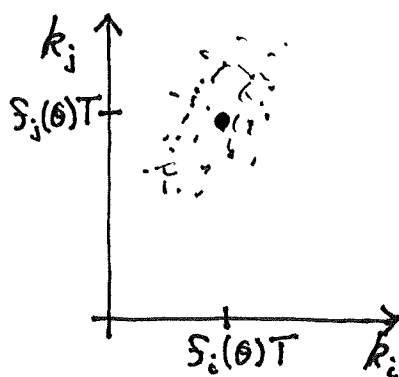
$$C_{ij} = \frac{\langle (k_i - \bar{k}_i(\theta)T)(k_j - \bar{k}_j(\theta)T) \rangle}{\sigma_i \sigma_j}$$

Pearson correlation coefficient

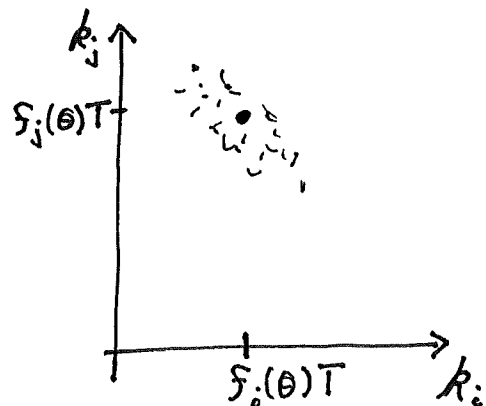
lower case c



Independence assumption
(Uncorrelated)



MT neurons
are positively
correlated



Negative
correlation

Shadlen and Newsome hypothesized in 1994 that neural correlations may be the root cause of perceptual errors. Suppose the monkey makes

its decision by pooling over left and right selective neurons in MT.

$$R = \frac{1}{N_R} \sum_{i=1}^{N_R} k_i^{(R)}, \quad L = \frac{1}{N_L} \sum_{i=1}^{N_L} k_i^{(L)}, \quad \begin{matrix} L > R \\ \text{or} \\ R > L \end{matrix} ?$$

What are the statistics of these pooled signals?

$$\langle R \rangle = \frac{1}{N_R} \sum_{i=1}^{N_R} \langle k_i^{(R)} \rangle = f_R(\theta) T$$

$$\text{Var}(R) = \left\langle \left(\frac{1}{N_R} \sum_{i=1}^{N_R} k_i^{(R)} - f_R(\theta) T \right)^2 \right\rangle$$

$$= \left\langle \left(\frac{1}{N_R} \sum_{i=1}^{N_R} \underbrace{(k_i^{(R)} - f_R(\theta) T)}_{\delta k_i^{(R)}} \right)^2 \right\rangle$$

$$= \frac{1}{N_R^2} \sum_{j=1}^{N_R} \sum_{i=1}^{N_R} \langle \delta k_i^{(R)} \delta k_j^{(R)} \rangle$$

$$= \frac{1}{N_R^2} \left(N_R \sigma_R^2 + N_R(N_R - 1) C_R \sigma_R^2 \right)$$

$$= C_R \sigma_R^2 + \frac{(1 - C_R) \sigma_R^2}{N_R}$$

$$\underbrace{\hspace{1cm}}_{\text{Nonzero if } C_R \neq 0} \quad \underbrace{\hspace{1cm}}_{\sim \text{zero for } N_R \gg 1}$$

Both R and L are noisy; a nonzero error rate remains.

Correlated Gaussian noise model of neural encoding

$$\underbrace{\begin{bmatrix} k_1 \\ \vdots \\ k_N \end{bmatrix}}_{\text{joint sampling}} \sim \mathcal{N} \left(\underbrace{\begin{bmatrix} f_1(\theta) T \\ \vdots \\ f_N(\theta) T \end{bmatrix}}_{\text{mean vector } \mu_i = f_i(\theta) T}, \underbrace{C(\theta)}_{\text{covariance matrix}} \right)$$

$$C_{ij} = \langle \delta k_i \delta k_j \rangle$$

$$P(k_1, \dots, k_N | \theta) = \frac{1}{\sqrt{(2\pi)^N \det C}} e^{-\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N (k_i - \mu_i) C^{-1}_{ij} (k_j - \mu_j)}$$

$\det(C)$: determinant of C

C^{-1}_{ij} : matrix inverse of C (i,j component)

The Fisher information of this model has been analyzed for many choices of C . Lessons:

1. $J(\theta)$ scales with N for some models and saturates for others. Saturation implies "information limiting correlations."
2. State-of-the art models only saturate at very high values, and accurate decoding of neural activity can exceed animal behavior.
3. The performance of suboptimal decoders can saturate, even if $J(\theta)$ scales. Example of uniform correlations

$$C_{ij} = \sigma^2 \delta_{ij} + C \sigma^2 (1 - \delta_{ij})$$

Here $J(\theta)$ scales, pooling saturates in performance (as argued before), optimal linear decoding does not. Behavioral limits may result from suboptimal decoders and correlated noise.

Biased decoding

Decoders can be biased.

$$b = \langle \hat{S} \rangle - S_0 = \underbrace{S_e - S_0}_{\text{generally nonzero}}$$

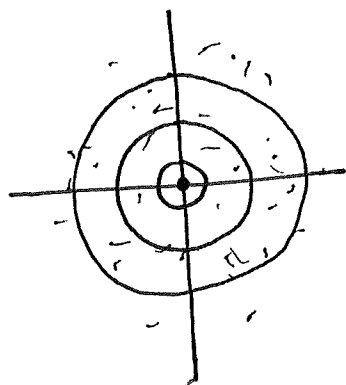
Bias-variance decomposition of error:

$$\underbrace{\langle (\hat{S} - S_0)^2 \rangle}_{\text{Mean squared error}} = \underbrace{\langle (\hat{S} - S_e + S_e - S_0)^2 \rangle}_{\text{crossterm averages to zero}} = \text{Var}(\hat{S}) + b^2$$

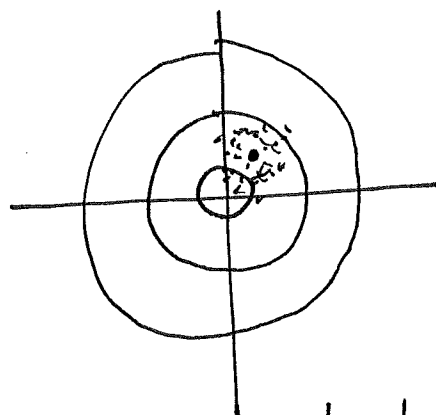
Bias can improve performance by reducing variance.

$$\text{Var}(\hat{S}) \geq \frac{(1 + db(s)/ds)^2}{J(s)}$$

Modified Cramer-Rao Lower Bound



High Var
No Bias



Low Var
Low Bias

The Bayesian decoder is an important example

$$P(s|n) = \frac{P(n|s)P(s)}{P(n)}$$

$$\begin{aligned} \hat{S} &= \arg \max_s \log P(s|n) \\ &= \arg \max_s \left(\underbrace{\ell(s)}_{\text{log-likelihood}} + \underbrace{\log P(s)}_{\text{prior probability}} \right) \end{aligned}$$

It is biased to favor stimuli that are a priori common. The prior "regularizes" the decoding problem.